EE204 Assignment 2

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Objective

To design, simulate, and test a three-op-amp instrumentation amplifier with a single-element gain control that varies the overall differential gain from 2 to 1000, as described in Example 2.3 of Chapter 2.

Theory

The instrumentation amplifier is a precision circuit composed of three op-amps: two non-inverting buffers forming the first stage and a differential amplifier forming the second.

1. In the first stage, each input signal is applied to a non-inverting amplifier. The two amplifiers share a common resistor $2R_1$between their inverting inputs, creating a high input resistance and enabling easy gain adjustment. The differential input voltage $v_{id} = v_{I2} - v_{I1}$produces an intermediate output difference.

$$V_{O2} - V_{O1} = \left(1 + \frac{R_2}{R_1}\right) V_{id}$$

Equation 1

2. The second stage is a precision difference amplifier with resistors $R_3$and R_4that reject common-mode signals and scale the differential output:

$$V_O = \frac{R_4}{R_3}(V_{O2} - V_{O1})$$

Equation 2

3. Combining both stages gives the total differential gain:

$$A_d = \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right)$$

Equation 3

But when $R_3 = R_4$ the second stage has a unity gain, so the overall gain depends on $R_1 \ and \ R_2$

$$A_d = 1 + \frac{R_2}{R_1}$$

Equation 4

The improved configuration (shared $2R_1$) prevents amplification of common-mode voltage in the first stage, thereby enhancing CMRR. Precise resistor matching in the second stage further increases common-mode rejection.

Design parameters

$$R_3 = R_4 = 10\ k\Omega, \qquad R_2 = 50\ k\Omega, \qquad 2R_1 = R_{1f} + R_{1v} = 100\Omega + 100k\Omega\ pot.$$

Equation 5

This allows a variable gain from 2 to 1000 while maintaining high input impedance and low common-mode gain.

Procedure

1. Build circuit in LTSpice using the three ideal op-amps powered by ±15V.
2. Apply a small differential input and vary $2R_1$ to observe gain changes.
3. Apply identical voltages to both inputs to verify minimal output under common mode conditions (high CMRR).
4. Implement the same circuit physically and compare the measured vs simulated results.

Results

The instrumentation amplifier circuit was successfully constructed in LTspice using three OP07 op-amps configured according to *Example 2.3* of Chapter 2. The schematic of the final design is shown in Figure 1with all the figures for resistors given above in design parameters

Schematic of the final design

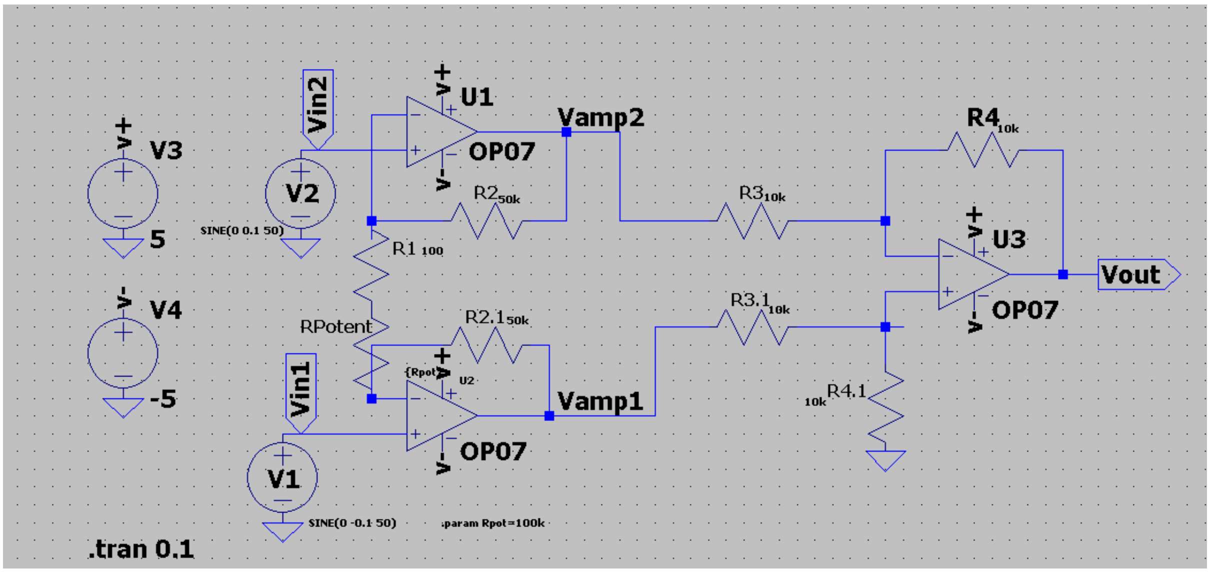

Figure 1

The LM348 quad-op-amp pinout used for the practical implementation is shown in Figure 3, and the physical IC orientation with the three active amplifiers (A1, A2, A3) labelled is displayed in Figure 2. These were used to correctly map power pins $V_{CC+}$and $V_{CC-}$and to identify which internal op-amps were configured for the buffer and differential stages

Pin Layout for LM348

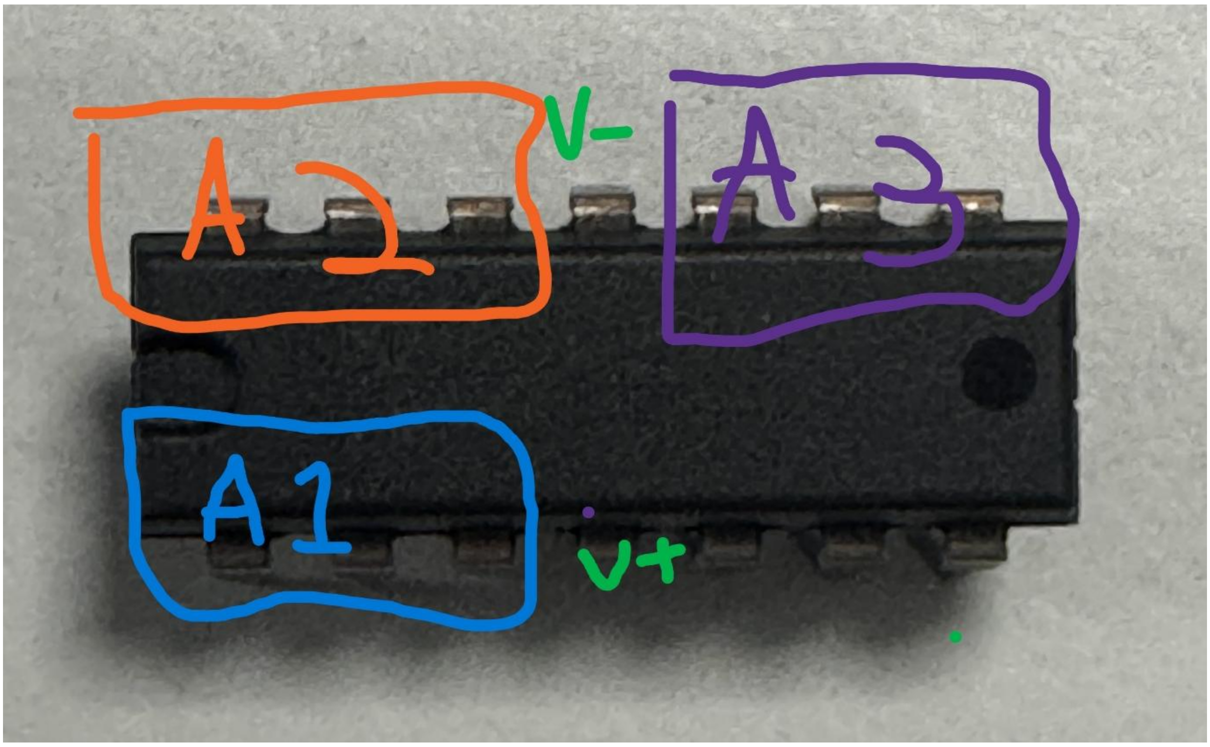

Figure 2

Datasheet For LM348

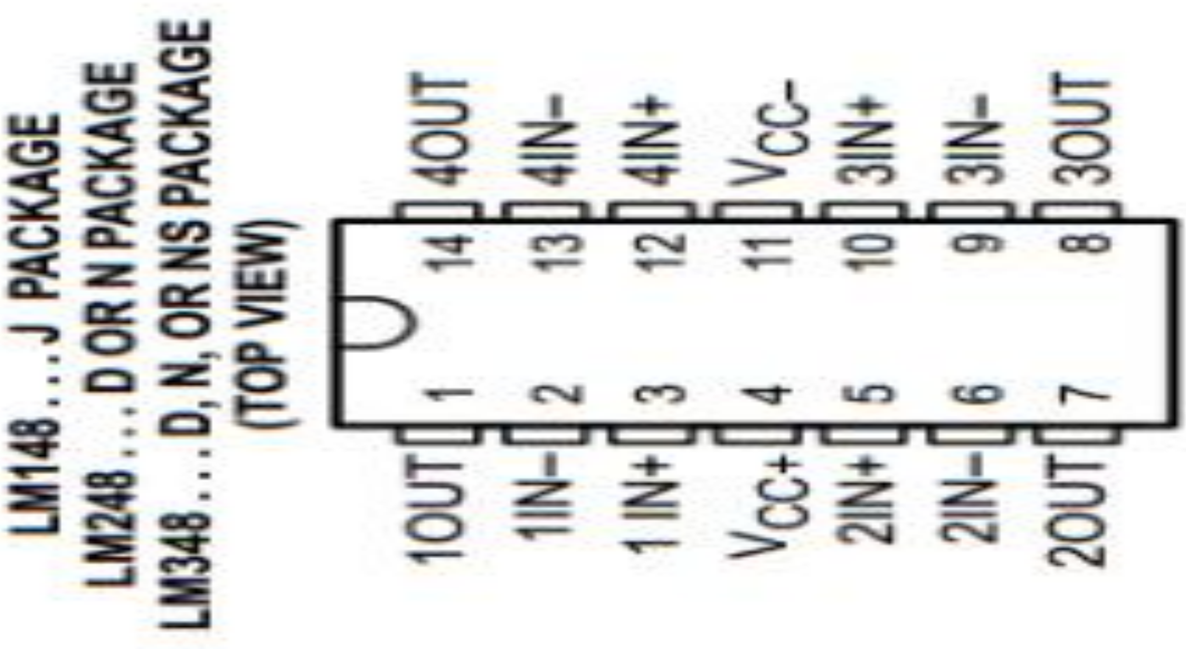

Figure 3

The circuit was then assembled as seen in Figure 4. using a breadboard and a LM348, however despite correct wiring and power supply connections the physical model didn't produce the expected amplified output. The waveform was unstable and didn't match the simulation results. Which wasn't resolved and would need further troubleshooting to verify hardware implementations.

Physical Implementation

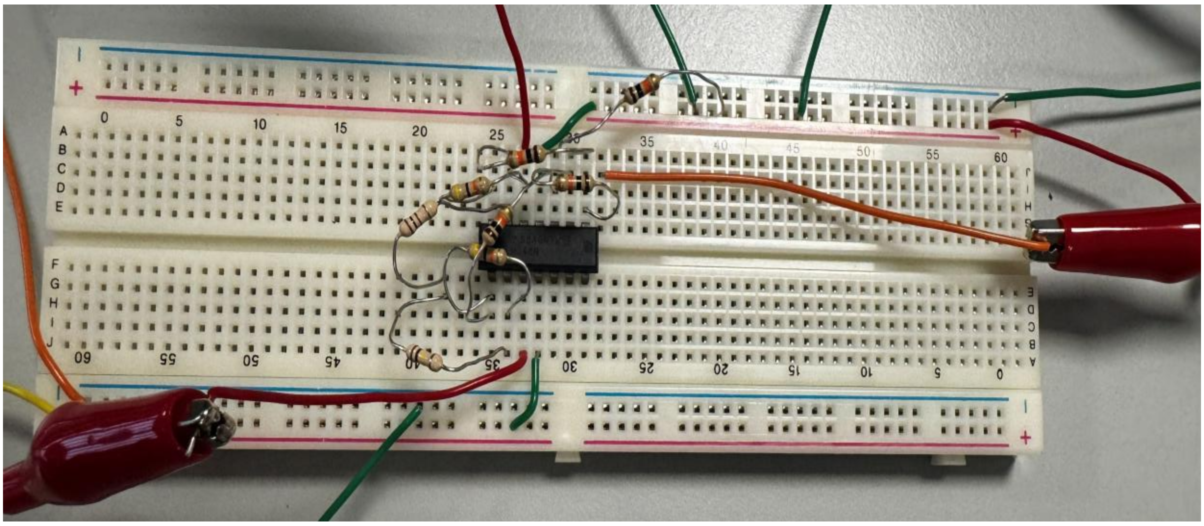

Figure 4

Transient analysis was then carried out in LTspice using a ±5 V dual supply and differential input signals $V_1 = \text{SINE}(0, -0.1, 50)$and $V_2 = \text{SINE}(0, 0.1, 50)$. The simulated waveforms for $R_{\text{pot}} = 100\ \text{k}\Omega$ are presented in **Error! Reference source not found.**. The intermediate outputs $V_{\text{amp1}}$and V_{amp2} follow their respective inputs $V_{\text{in1}}$and V_{in2}in phase, confirming that both input amplifiers operate in the non-inverting mode. The final output voltage V_{out}(red trace) is clearly amplified and in phase with V_{in2} while inverted relative to V_{in1}, which agrees with the expected behaviour of the difference-amplifier stage. The measured amplitude was approximately twice the differential input amplitude, giving an overall gain $A_d \approx 2\ given\ R_{pot} = 100k\Omega$ as predicted by the theoretical expression.

LtSpice Simulation Results

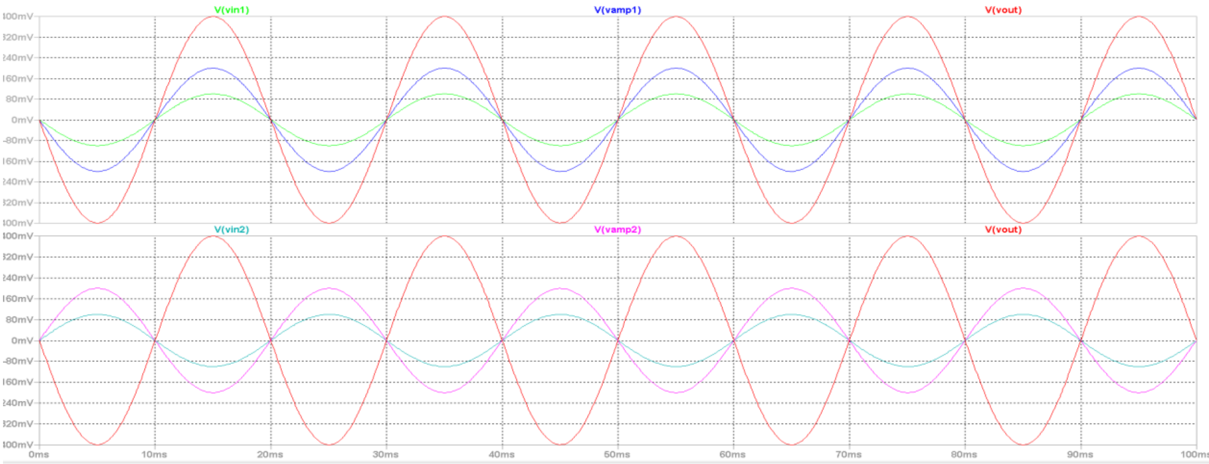

Figure 5

Discussion

The simulated results confirmed that the three-op-amp instrumentation amplifier operated as expected, providing stable differential amplification with high input impedance and good common-mode rejection. The adjustable resistor network $2R_1 = 100\Omega + R_{\text{pot}}$successfully varied the overall gain, demonstrating the single-element control described in *Example 2.3* (pp. 94–97). The phase relationships observed in Figure 5 verified correct non-inverting operation of the first stage and proper subtraction in the difference amplifier.

However, the physical implementation did not produce the expected output. This discrepancy was likely caused by practical issues such as loose breadboard connections, incorrect pin orientation on the LM148 IC, or power-supply noise. These factors can easily affect offset voltages and biasing, preventing the circuit from operating within the linear region. Additional troubleshooting, including verifying rail voltages, rechecking wiring, and ensuring the common resistor link is secure, would be necessary to replicate the simulated behaviour in hardware.

Conclusion

The instrumentation amplifier was successfully designed and verified through LTspice simulation, showing a clear differential gain of approximately two for $R_{\text{pot}} = 100\text{k}\Omega$and proportional gain increase as R_{pot}decreased. The simulation results aligned well with the theoretical predictions from *Example 2.3*, confirming that the design provides accurate differential amplification with high input resistance and effective common-mode rejection. Although the physical model did not yield the expected results, the experiment demonstrated a solid understanding of the amplifier's operation and highlighted the importance of careful hardware construction and troubleshooting in achieving reliable practical performance.